

HealthFirst Digital Dental Camp Platform

Product Requirements Document (PRD)

1. Project Overview

HealthFirst plans to build a digital platform to manage dental screening camps conducted in corporates, apartments, factories, and institutions. The system will streamline patient registration, enable dentists to perform rapid dental screenings, generate digital dental reports, and produce analytics dashboards for organizations. The platform must support high-volume camps (300–800 participants per day) while maintaining a fast, mobile-friendly interface.

2. Objectives

- Digitize the dental camp workflow.
- Allow fast patient registration and queue management.
- Provide dentists with a rapid screening interface.
- Automatically generate digital dental reports.
- Provide analytics dashboards for corporate wellness programs.
- Enable scalable deployment across multiple locations.

3. User Roles

- Admin – manages camps, users, and analytics dashboards.
- Dentist – performs dental screening and generates reports.
- Assistant – helps with patient registration and queue management.
- Participant – registers for screening and receives dental report.

4. Patient Registration Module

- Mobile-friendly registration form.
- Fields: Name, Age, Gender, Phone Number, Organization/Apartment/Company name.
- Optional fields: Department, Employee ID, Flat number.
- System generates unique Patient ID automatically.
- Registered participants appear in screening queue.

5. Doctor Screening Module

- Dentist login dashboard.
- Patient queue with search functionality.
- Quick screening interface with minimal typing.
- Fields: Oral Hygiene, Caries, Gingivitis, Malocclusion, Notes.
- Treatment recommendation options: Scaling, Filling, Extraction, RCT, Orthodontic consultation.

6. Digital Dental Report Module

- Automatic PDF report generation after screening.
- Report includes patient details, screening findings, treatment advice, and dentist name.
- Reports stored securely in database.
- Reports accessible through admin panel.

7. Camp Analytics Dashboard

- Display metrics such as total participants, caries prevalence, gingivitis prevalence.
- Generate summary oral health reports for corporates or organizations.
- Export analytics data to Excel or PDF.

8. System Workflow

- Participant registers for screening.
- Participant added to screening queue.
- Dentist performs screening and records findings.
- System generates digital dental report.
- Analytics dashboard updates automatically.

9. Suggested Technology Stack

- Frontend: React or Next.js.
- Backend: Node.js or Django.
- Database: PostgreSQL or Firebase.
- PDF generation: PDFKit or Puppeteer.
- Hosting: AWS, Google Cloud, or Vercel.

10. Future AI Enhancements

- AI report writing to summarize screening findings.
- AI risk scoring for dental disease prediction.
- AI chatbot for patient education.
- Intraoral image analysis for detecting dental issues.